Arash Jalil Khabbazi, MS

PhD Student, Mechanical Engineering, Purdue University

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Profile

I work on machine learning, optimization, and advanced control algorithms to make buildings and energy systems smarter and more efficient. I bring experience from both research and industry to solving practical engineering problems.

Education

Purdue University, IN, United States — 2023 –
PhD in Mechanical Engineering, Minor in Computational Science and Engineering — **GPA:** 4.0/4.0
(Adviser: Kevin J. Kircher)

University of British Columbia, BC, Canada — 2021 – 2023
MS in Mechanical Engineering — **GPA:** 4.33/4.33 (94%)
Thesis: *Mixing gaseous hydrogen into natural gas distribution pipelines* (Adviser: Sunny R. Li)

University of Tabriz, EA, Iran — 2016 – 2020
BS in Mechanical Engineering — **GPA:** 4.0/4.0 (19.12/20) — **Rank:** 1/155+
Thesis: *Thermodynamic analysis of Kalina cycle system 11* (Adviser: S. Mohammad S. Mahmoudi)

Industry experience

Summer Intern, Tesla (Reno, NV, United States) — 05/2025 – 08/2025

- Developed supervised ML algorithms and supported deployment of a thermal imaging system for real-time quality assessment for real-time quality assessment, eliminating destructive lab tests.
- Utilized MySQL for data extraction and management, performing feature selection analyses (Python, JMP) to identify high-signal inputs for ML model development.
- Collaborated with Inverter Engineering and Advanced Manufacturing teams to align models with ongoing projects, with deployment strategies projected to save Tesla over $1M annually in testing costs.

Student Researcher, FortisBC (Vancouver, BC, Canada) — 09/2021 – 09/2023

- Supported FortisBC's technical assessments on hydrogen injection into natural gas pipelines and other renewable gas projects, including the UBC H_2Lab.
- Participated in monthly meetings with FortisBC engineers and research teams (materials, sensors, AI, safety) to review project findings and assess industry impacts.

Research experience

PhD Research Assistant, Purdue University — 09/2023 –

- Supported commissioning of Herrick Laboratories as a testbed for advanced HVAC control, contributing to BAS retrofits (Niagara/Modbus) and developing occupant-facing dashboards for real-time comfort and energy feedback.
- Built ML models for occupancy detection using CO_2 sensor data with 98.3% accuracy and automated pipelines (Python, InfluxDB, Linux) for real-time energy management in smart HVAC control.
- Designed and simulated HVAC control strategies with MPC and DeePC (CVX, CVXPY), showing DeePC matched MPC performance without requiring a system model, while addressing energy efficiency and occupant comfort.
- Conducted a large-scale review of 100+ publications (200+ field tests) on MPC and RL for residential and commercial HVAC, leading to a published review paper in Applied Energy.
- Modeled thermodynamic performance of low-GWP refrigerants in heat pump systems for cold climates using EES, analyzing efficiency trade-offs to support sustainable HVAC design and decarbonization.

MS Research Assistant, University of British Columbia — 07/2021 – 09/2023

- Led research on hydrogen injection into natural gas pipelines using CFD, real-gas modeling (C), and CAD-based pipeline design.
- Analyzed over 1 TB of simulation data, resulting in a journal paper, three conference presentations, and earning two conference awards.

Skills

Programming & Scripting: Python, C, MATLAB, R, Git, Markdown, HTML

ML & Data Science: TensorFlow, Keras, scikit-learn, XGBoost, NumPy, Pandas, SciPy, Matplotlib, Seaborn

Data Systems & Operating Environments: MySQL, InfluxDB, Grafana, Linux

Control & Communication Protocols: Modbus, IO-Link, MTConnect

Engineering & Simulation Tools: EES, ANSYS Workbench, OpenFOAM, Tecplot, SOLIDWORKS, CATIA, JMP

Publications

Journal articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review," *Applied Energy*, 2025.

1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions," *International Journal of Hydrogen Energy*, 2024.

Conference proceedings

6. **A.J. Khabbazi** and K.J. Kircher, "Do Small HVAC Control Demonstrations in Larger Buildings Risk Overestimating Savings?," *under review*, 2025.

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5. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, "Active current limiting control of residential appliances for breaker panel protection across the US: a parametric study," *CISBAT*, 2025, pp. 1–6.

4. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, K.J. Kircher, "What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?," *International High Performance Buildings Conference*, 2024, pp. 1–10.

3. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, "Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions," *Canadian Society for Mechanical Engineering International Congress*, 2023, pp. 1–6.

2. **A. Khabbazi**, R. Li, and J. Quinn, "Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid," *Canadian Society for Mechanical Engineering International Congress*, 2022, pp. 1–1.

1. **A. Khabbazi**, R. Li, and J. Quinn, "The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines," *International Green Energy Conference (IGEC-XIII)*, 2021, pp. 1–1.

Selected Honors & Awards

- ASHRAE Graduate Student Grant-in-Aid Award 2025
- Best Paper Award at CSME 2023 Conference 2023
- Best Presentation Award at CSME 2022 Conference, Advanced Energy Symposium 2022
- UBC Graduate Scholarship 2022
- UBC Dean's Entrance Scholarship 2021

Selected courses

Applied Mathematics & Data Science: Artificial Intelligence (ECE570) — Applied Machine Learning (ENGR418) — Statistical Methods (STAT511) — Advanced Mathematics I (MA527) — Numerical Computations

Automation, Control & IoT: Reinforcement Learning and Control (IE690) — Reinforcement Learning (CS593) — Intro to Convex Optimization (AAE561) — Industrial IoT Implementation (ME597)

Thermal & Energy Systems: Distributed Energy Resources (ME597) — Analysis of Thermal Systems (ME518) — Advanced Thermodynamics (ME500) — Refrigeration Systems — Power Plants — Heat Transfer